List

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List = Collection of elements
- numbers
- character
- dates
- list (collection its)

3 4 5

$x = 2$

$x = [2, 3, 4, 5, 6]$

like list → tuple, sets, dictionaries

x → mutable → modify/change the value in the List

$x = ['red', 'green', 'blue']$

range:- write numbers from starting numbers ending number

range(1, 10)

list:- it will collect all numbers and make a list

$x = list(range(1, 10))$

"abcd"

$x = list("abcd")$ → $['a', 'b', 'c', 'd']$

Index of list

$x = [1, 2, 3, 4, 5, 6, 7]$

Index of list: 0 1 2 3 4 5 6

$x = [1, 2, 3, 'Hi', True, [4, 5, 6]]$

0 1 2 3 4 5

$x[5] \Rightarrow [4, 5, 6]$

$x[5][0] \Rightarrow 4$

Built in functions in List

1) append() $x = [1, 2, 3, 4]$

$x.append(7)$

2) extend() $y = [8, 9]$

$x.extend(y)$ → $[8, 9]$

2) extend() y = [8, 9]

x.extend(y)

[1, 2, 3, 4, 7, 8, 9]

3) pop() → remove the last value from list

x.pop()

4) remove() → x.remove(8)

5) del() → del x[2]

del x → harmful

memory reference

y = x

list methods

x = [1, 2, 3, 4]

y = x

0 1 2 3

y = [1, 2, 3, 4] → 500

y[3] = 500

y = [1, 2, 3, 500]

x = [1, 2, 3, 500]

import copy

copy()

y = copy.copy(x)

y[3] = 600 x[ ]

deepcopy()

x = [1, 2, 3, [4, 5, 6]]

w = cp.deepcopy(x)

w[3][0] = 400

w will change

x will not change

clear()

x = [1, 2, 3]

x.clear()

[]

len()

count() → no. of times the value appears in the list

x = [1, 2, 3]

x.count(3)

y = [1, 2, 2, 3, 4, 2]

y.count(2)

→ 3

count() x = [...]

x.count(3)
= 1

y.count(2)
=> 3

index() x = [1, 2, 3]
0 1 2

x.index(2)
=> 1

x = [1, 2, 3, 4]
= [4, 3, 2, 1]
x[::-1]

reverse() reverse also reversing elements but it will modify list

sort()
ascending -> x.sort()
descending -> x.sort(reverse=True)

insert() x = [1, 2, 3, 4]
0 1 2

x.insert(2, 22)
x = [1, 2, 22, 3, 4]

dir() :- all methods available for List / tuple / dict.

dir(x)

help() :- description of method function(x) x.Method()
help(x.index)